Chapter 17 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

E. 1. Consider a solution consisting of the following two buffer systems:

$H_2CO_3 \rightleftharpoons HCO_3^- + H^+$ $\quad pK_a = 6.4$ $\quad K_a = 10^{-6.4}$

$H_2PO_4^- \rightleftharpoons HPO_4^{2-} + H^+$ $\quad pK_a = 7.2$ $\quad -\log K_a = 7.2$ $\quad K_a = 10^{-7.2}$

At pH 6.4 which one of the following is true of the relative amounts of acid and conjugate base present? $[H] = 10^{-6.4}$

A) $[H_2CO_3] > [HCO_3^-]$ and $[H_2PO_4^-] > [HPO_4^{2-}]$

B) $[HCO_3^-] > [H_2CO_3]$ and $[HPO_4^{2-}] > [H_2PO_4^-]$

C) $[H_2CO_3] > [HCO_3^-]$ and $[HPO_4^{2-}] > [H_2PO_4^-]$

D) $[H_2CO_3] = [HCO_3^-]$ and $[HPO_4^{2-}] > [H_2PO_4^-]$

E) $[H_2CO_3] = [HCO_3^-]$ and $[H_2PO_4^-] > [HPO_4^{2-}]$

B. 2. How much solid NaCN must be added to 1.0 L of a 0.5 *M* HCN solution to produce a solution with pH 7.0? $K_a = 6.2 \times 10^{-10}$ for HCN.

A) 0.0034 g B) 0.15 g C) 160 g D) 24 g E) 11 g

C. 3. If 16 mL of 0.78 *M* HCl is added to 109 mL of 0.20 *M* NaOH, what is the final pH?

A) 0.76 B) 13.24 C) 12.87 D) 7.00 E) 1.13

A. 4. The salt AgCl is _____ soluble in strong acid solution than in water.

A) neither more nor less (that is, AgCl is equally soluble in strong acid and in water)

B) more

C) The K_{sp} value for AgCl must be known to answer this question.

D) less

$[Ag^+]\cdot[Cl^-]$ key factor.

B. 5. A 200.0-mL sample of the weak acid H_3A (0.100 *M*) is titrated with 0.200 *M* NaOH. After 200.0 mL NaOH is added, what is the major species in the titration? (Water is always assumed to be a major species.) 0.02 mol 0.04 mol.

A) HA^{2-}, A^{3-} B) HA^{2-} C) H_2A^- D) H_2A^-, HA^{2-} E) A^{3-}

E 6. Silver acetate ($AgC_2H_3O_2$) is a sparingly soluble salt with $K_{sp} = 1.9 \times 10^{-3}$. Consider a saturated solution in equilibrium with the solid salt. Compare the effects on the solubility of adding to the solution either the acid HNO_3 or the base NH_3.

A) NH_3 would increase the solubility, but HNO_3 would decrease it.

B) NH_3 would decrease the solubility, but HNO_3 would increase it.

C) NH_3 would increase the solubility, but HNO_3 would have virtually no effect.

D) Either substance would decrease the solubility.

E) Either substance would increase the solubility.

E. 7. Calculate the pH at the equivalence point for the titration of 1.0 *M* ethylamine, $C_2H_5NH_2$, by 1.0 *M* perchloric acid, $HClO_4$. (pK_b for $C_2H_5NH_2$ = 3.25)

A) 6.05 B) 2.24 C) 2.09 D) 5.38 E) 5.53

A. 8. What is the pH of a solution that results when 0.012 mol HNO_3 (H^+) is added to 610.0 mL of a solution that is 0.26 *M* in aqueous ammonia ($NH_3 \cdot H_2O$) and 0.46*M* in ammonium nitrate (NH_4NO_3). Assume no volume change. (K_b for $NH_3 = 1.8 \times 10^{-5}$.)

A) 8.96 B) 9.56 C) 5.04 D) 9.01 E) 4.44

Chapter 17 Homework

C. 9. Which of the following compounds has the lowest solubility, in moles per liter, in water?

A) MgC_2O_4 $K_{sp} = 8.6 \times 10^{-5}$
B) $PbSO_4$ $K_{sp} = 1.3 \times 10^{-8}$
C) CdS $K_{sp} = 1.0 \times 10^{-28}$
D) $Al(OH)_3$ $K_{sp} = 2 \times 10^{-32}$
E) $Sn(OH)_2$ $K_{sp} = 3 \times 10^{-27}$

还要比较溶解度 S.

$K_a \cdot K_b = \frac{[NH_3][H^+]}{[NH_4^+]} \cdot \frac{[NH_4^+][OH^-]}{[NH_3]} = K_w$

B 10. You are given 5.00 mL of an H_2SO_4 solution of unknown concentration. You divide the 5.00-mL sample into five 1.00-mL samples and titrate each separately with 0.1000 *M* NaOH. In each titration, the H_2SO_4 is completely neutralized. The average volume of NaOH solution used to reach the endpoint is 15.3 mL. What was the concentration of H_2SO_4 in the 5.00-mL sample?

A) 0.306 *M*
B) 0.765 *M*
C) 3.06 *M*
D) 1.53×10^{-3} *M*
E) 1.53 *M*

$NH_4^+ + H_2O \rightleftharpoons NH_3 \cdot H_2O + H^+$

$K_a(NH_4^+) \cdot K_b(NH_3) = K_w$

$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

① $pK_a \approx pH$. ② [base] ≈ [acid]

D. 11. Which of the following solutions will be the **best** buffer at a pH of 9.26? (K_a for CH_3COOH is 1.8×10^{-5}; K_b for NH_3 is 1.8×10^{-5}.)

A) 0.20 *M* NH_3 and 0.20 *M* NH_4Cl
B) 5.0 *M* CH_3COOH and 5.0 *M* NH_4Cl
C) 0.20 *M* CH_3COOH and 0.20 *M* CH_3COONa
D) 5.0 *M* NH_3 and 5.0 *M* NH_4Cl
E) 5.0 *M* CH_3COOH and 5.0 *M* NH_3

$K_a = \frac{[H^+][A^-]}{[HA]}$

$-\log K_a = -\log [H^+] - \log \frac{[A^-]}{[HA]}$ $\frac{[base]}{[acid]}$

pK_a = pH

$pK_a = pH - \log \frac{[base]}{[acid]}$

D. 12. You have a solution of 0.10 *M* Cl^- and 0.10 *M* CrO_4^{2-}. You add 0.10 *M* silver nitrate dropwise into the solution. K_{sp} for Ag_2CrO_4 is 9.0×10^{-12} and for AgCl is 1.6×10^{-10}. Which of the following will precipitate first?

10^{-9} 10^{-6}.

A) silver nitrate
B) silver chromate
C) cannot be determined from the information given
D) silver chloride
E) none of these

B. 13. Equal volumes of 0.1 *M* HCl and 0.1 *M* CH_3COOH are titrated with 0.1 *M* NaOH. Which of the following would be equal for both titrations?

A) the pH at the equivalence point
B) the volume of NaOH added to reach the equivalence point
C) the initial pH
D) the pH at the halfway point
E) two of the above

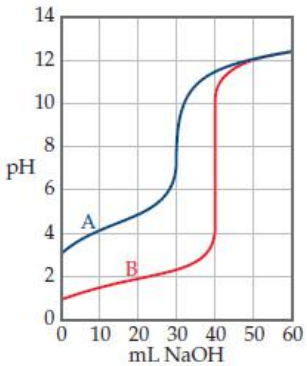
Part B Short questions (12 points)

16. (4 points) The accompanying graph shows the titration curves for two monoprotic acids.

(a) Which curve is that of a strong acid?

(b) What is the approximate pH at the equivalence point of each titration?

(c) 40.0 mL of each acid was titrated with 0.100 *M* base. Which acid is more concentrated?



17. (8 points) The solubility of $CaCO_3$ is pH dependent.

(a) Calculate the molar solubility of $CaCO_3$ ($K_{sp} = 4.5 \times 10^{-9}$) neglecting the acid–base character of the carbonate ion.

(b) Use the K_b expression for the CO_3^{2-} ion (K_{a2} for H_2CO_3 is 5.6×10^{-11}) to calculate the equilibrium constant for the reaction

$$CaCO_3(s) + H_2O(l) \rightleftharpoons Ca^{2+}(aq) + HCO_3^-(aq) + OH^-(aq)$$

(c) If we assume that the only sources of Ca^{2+}, CO_3^{2-}, and OH^- ions are from the dissolution of $CaCO_3$, what is the molar solubility of $CaCO_3$ using the preceding expression? What is the pH?

(d) If the pH is buffered at 7.5, what is the molar solubility of $CaCO_3$?

Chapter 17 Homework Answer Sheet

Name: 陈.佳 **Student ID:** 12311410 **Instructor:** **Score:**

Part A Multiple choice (13 points)

1-5 EBCAB

6-10 EEACB.

11-13 DDB.

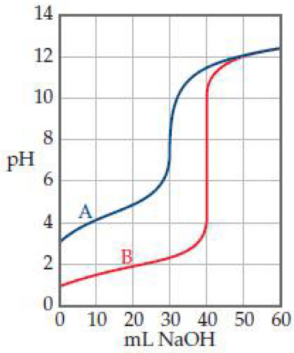
Part B Short questions (12 points)

16. (4 points) The accompanying graph shows the titration curves for two monoprotic acids.

(a) Which curve is that of a strong acid?

(b) What is the approximate pH at the equivalence point of each titration?

(c) 40.0 mL of each acid was titrated with 0.100 *M* base. Which acid is more concentrated?



(a) B.

(b) A: 8

B: 7.

(c) B.

17. (8 points) The solubility of $CaCO_3$ is pH dependent.

(a) Calculate the molar solubility of $CaCO_3$ ($K_{sp} = 4.5 \times 10^{-9}$) neglecting the acid–base character of the carbonate ion.

(b) Use the K_b expression for the CO_3^{2-} ion (K_{a2} for H_2CO_3 is 5.6×10^{-11}) to calculate the equilibrium constant for the reaction

$$CaCO_3(s) + H_2O(l) \rightleftharpoons Ca^{2+}(aq) + HCO_3^-(aq) + OH^-(aq)$$

(c) If we assume that the only sources of Ca^{2+}, CO_3^{2-}, and OH^- ions are from the dissolution of $CaCO_3$, what is the molar solubility of $CaCO_3$ using the preceding expression? What is the pH?

(d) If the pH is buffered at 7.5, what is the molar solubility of $CaCO_3$?

(a) $S = \sqrt{K_{sp}} = 6.7 \times 10^{-6} M.$

(b). $K = [Ca^{2+}][HCO_3^-][OH^-] = \dfrac{[Ca^{2+}][CO_3^{2-}][HCO_3^-][OH^-][H^+]}{[CO_3^{2-}][H^+]}$

$$= \frac{K_{sp} \cdot K_w}{K_{a2}} = 8.0 \times 10^{-13}.$$

(c). $S = \sqrt[3]{K} = 9.3 \times 10^{-5} M.$

(d). $[H^+] = 3.16 \times 10^{-8} M.$

$$\frac{[CO_3^{2-}]}{[HCO_3^-]} = \frac{K_{a2}}{[H^+]} \Rightarrow \frac{[HCO_3^-]}{[CO_3^{2-}]} = 560.$$

$$\Rightarrow \underbrace{S}_{Ca^{2+}} \cdot \underbrace{\frac{S}{560}}_{CO_3^{2-}} = K_{sp} \Rightarrow S = 1.6 \times 10^{-3} M.$$